STAT 415/615 **Fall 2019**

Regression

Course syllabus

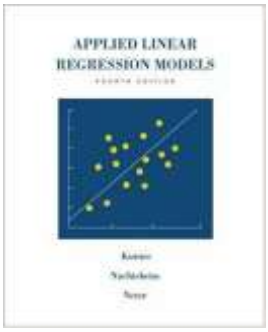
Time	Monday & Thursday 8:20 – 9:35 pm
Room	Kerwin Hall, ST-01
Instructor	Michael Baron
Office	DMTI 106-D
Office hours	Monday & Thursday 7:00 – 8:00 pm
Phone	202-885-3130
Email	baron@american.edu

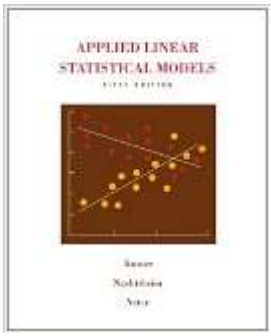
Textbook

- *Applied Linear Regression Models*, 4th edition, by M. Kutner, C. Nachtsheim, and J. Neter. McGraw-Hill, 2004 (701 pp). ISBN 0073014664

  OR

- *Applied Linear Statistical Models*, 5th edition, by M. Kutner, C. Nachtsheim, J. Neter, and W. Li. McGraw-Hill, 2004 (1396 pp). ISBN 007310874X





BUT NOT BOTH. Chapters 1-14 are identical in these books.

Text only, without a CD, is fine. All textbook data sets are available on our Blackboard course page.

Course description:

Simple and multiple regression, least squares, curve fitting, graphic techniques, and tests and confidence intervals for regression coefficients. Prerequisite: STAT-302, STAT-320, or STAT-514/614.

Course plan:

1. Introduction, motivation, and examples [1.1-1.2]
2. Review of elementary statistical inference. Normal, t, and F distributions. Hypothesis testing. P-values. [Handouts]
3. Introduction to R and SAS [Handouts]
4. Linear regression: model, estimation, inference, prediction. Regression and correlation. R^2 [Chap. 1-2]
5. Regression diagnostics: non-normality, nonlinearity, heteroscedasticity [Chap. 3]
6. Simultaneous estimation. Other regression models [Chap. 4]
7. Multiple regression. Matrix approach (Stat-615). Analysis of variance. Analysis of residuals. Partial correlation and multiple correlation coefficient [Chap. 5-6]
8. Model building. Model selection and validation. Extra sum of squares [Chap. 7-8]
9. Regression diagnostics-II. Influential observations and outliers. Effect of multicollinearity. Robust regression. Ridge regression [Chap. 9-10]
10. Regression diagnostics-III. Symptoms and remedies. Transformation of variables. Missing data. Analysis of covariance. Comparison of regression lines. [Chap. 10]
11. Dummy variables and related methods [Chap. 11]
12. (If time permits) Nonlinear relations. Logistic regression. [Chap. 13-14]

Learning Objectives (Acknowledgement: Thank you, Professor Jun Lu!)

The learning objective of this course is to give you the main concepts and a working knowledge of regression techniques that are routinely used to analyze different types of data. At the end of this course, you are expected to be able to:

- Identify studies and data sets where regression can be used to address the questions of interest.
- Use software to graphically display regression data.
- Propose a regression model to address the research questions in a study.
- Understand the principle of the Least Squares Estimation.
- Use software to conduct regression analysis. This includes variable selection, parameter estimation, diagnostics, and prediction.
- Interpret and summarize the results of regression analysis results in the context of the study.
- Understand limitations of the regression analysis.
- Design and conduct a study to investigate a research problem using real-world data and regression analysis. (Stat 615)
- Derive the least squares estimators for linear regression. (Stat 615)
- Write the linear regression model in matrix form. (Stat 615)
- Understand matrix derivations for estimation, testing, and model building in multiple linear regression. (Stat 615)

Assignments and Grading:

Weekly homework and mini-projects	20%	Homework will be assigned weekly and graded quickly. A steady effort to work out all the assigned problems is essential for learning statistical methods and for the successful performance in this course. Homework solutions will be given. A typical homework will include a few "toy" problems to do by hand, to see how things work, and a few computer projects - applied problems with realistic data sets to do using computer software. Collaboration on a homework is ok. Even encouraged! Quizzes and exams are to be done individually.
Weekly quizzes	25%	10- to 15-minute quizzes in the end of every Thursday class. Each quiz covers the material of the preceding week and the latest homework. During a quiz, you may use **one cheat sheet**.
Midterm Test	25%	The midterm covers several chapters of the material. Taken in class; **notes and our Blackboard course materials are allowed**. Time is limited – 1 hour 15 minutes.
Final Test Dec 9, 8:20-10:20 pm	30%	The final covers the 2nd part of the course, but it is cumulative indirectly because the 2nd part is heavily based on the 1st part. Taken in class; **notes and our Blackboard course materials are allowed,** but of course, serious preparation is essential for getting a good grade. Time: 2 hours.

90 – 100 % = A	87 – 90 % = A-	83 – 87 % = B+	80 – 83 % = B
77 – 80 % = B-	73 – 77 = C+	70 – 73 % = C	60 – 70 % = C-

Software: During the course, we'll be learning regression methods and implementing them in **R** and **SAS**. Both languages will be introduced practically from scratch. Classroom demonstrations and examples will be in R. SAS will be presented in handouts.
For all computer assignments, use the language of your choice. Advanced programming skills and advanced computer knowledge are *not* required.

To use **R**, install it from https://cran.r-project.org/, free of charge.

SAS licenses are generally expensive, however, AU students have several free solutions:

- The latest release of SAS is installed in some computer labs on campus - CTRL lab, Anderson Computing Cluster, Kreeger, SPA lab
- SAS is available through the Virtual Computing Lab (VCL) - www.american.edu/vcl/
- If you have an AU owned computer, you can install SAS on it - 202-885-3862, rsg@american.edu
- The University Edition of SAS is free and offline - www.sas.com/en us/software/university-edition.html
- SAS On Demand is online, via cloud, and also free - www.sas.com/en_us/software/on-demand-for-academics.html

  Overall support of SAS for AU community - www.american.edu/ctrl/software sas.cfm

Support Services. A wide range of services is available to support you in your efforts to meet the course requirements.

- *Mathematics & Statistics Tutoring Lab* (x3154, x3120, Don Myers Building, Room 103) provides tutoring in Mathematics and Statistics. Lab hours are Mo-Th 11 am – 8 pm, Fr 11 am – 3 pm, and Su 3 pm – 8 pm. http://www.american.edu/cas/mathstat/tutoring.cfm
- *CTRL Connect* – software support with R and CTRL workshops (ctrl@american.edu, x2117)
- *Academic Support and Access Center* (x3360, MGC 243) offers study skills workshops, individual instruction, tutor referrals, writing support, and technical and practical support and assistance with accommodations for students with physical, medical, or psychological disabilities. Open Mon-Thu 9:00 am – 7:00 pm and Fri 9:00 am – 5:00 pm. https://www.american.edu/provost/academic-access/

  Writing support is also available in the Writing Center, Battelle-Tompkins 228.
- *Counseling Center* (x3500, MGC 214) offers counseling and consultations regarding personal concerns, self-help information, and connections to off-campus mental health resources. https://www.american.edu/ocl/counseling/

Emergency Preparedness. In the event of an emergency, students should refer to the AU Web site (http://www.american.edu/emergency) and the AU information line at (202) 885-1100 for general university-wide information. In case of a prolonged closure of the University, I send updates to you by email and will post all announcements on the course web site.

Academic Integrity. According to the University's Academic Integrity Code, I am required to report cases of *academic dishonesty* to the Dean of the College of Arts and Sciences. AU's academic code is found at http://www.american.edu/academics/integrity/

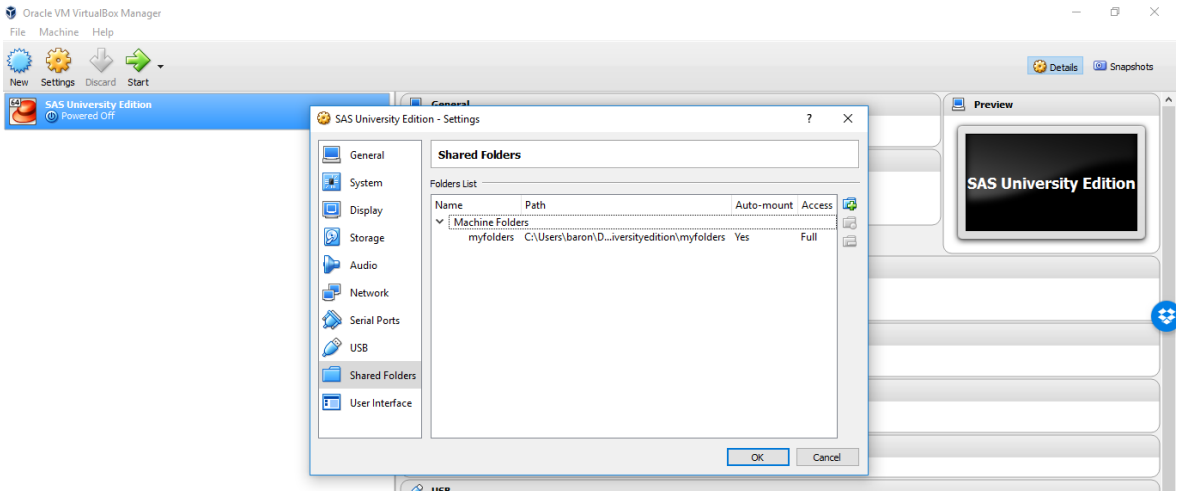
STAT 615/415 **Regression**

SAS University Edition: Shared Folder for Data Files and SAS Codes

In the University Edition of SAS, you can keep data files in a designated **shared folder.** This folder is local on your computer, and it is accessible by SAS.

1. **See if you have a shared folder.**

Start **Oracle VM VirtualBox Manager**. Go to Settings, and in the Settings window, choose Shared Folders

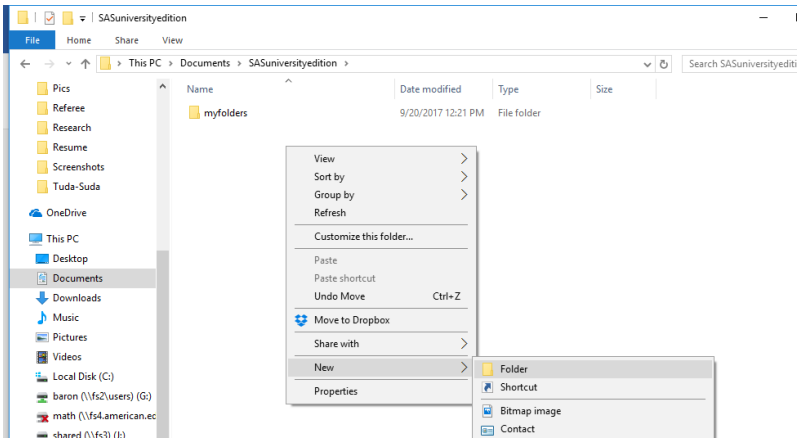


If there is a folder under "Machine Folders", then this is your shared folder. Locate it on your local computer. You may use it for file transfer between you and SAS.

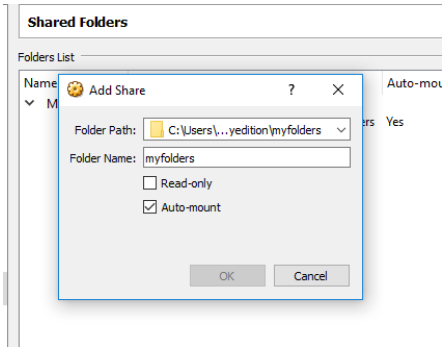
2. **Set up a shared folder.**

If there is nothing below "Machine Folders", we'll have to set up your shared folder.

Within your "My Documents", create a folder "SASuniversityedition", open it, and create a subfolder "myfolders". To create a folder, right-click your mouse and choose New -> Folder.



Then, back in **Oracle VM VirtualBox Manager**'s Settings window, click on a green plus on the right margin, next to the word Access. In the new Add Share window, copy the whole path of your "myfolders" folder.
Check the "Auto-mount" box and uncheck the "Read Only" box.
Click OK, OK, and your shared folder is all set.

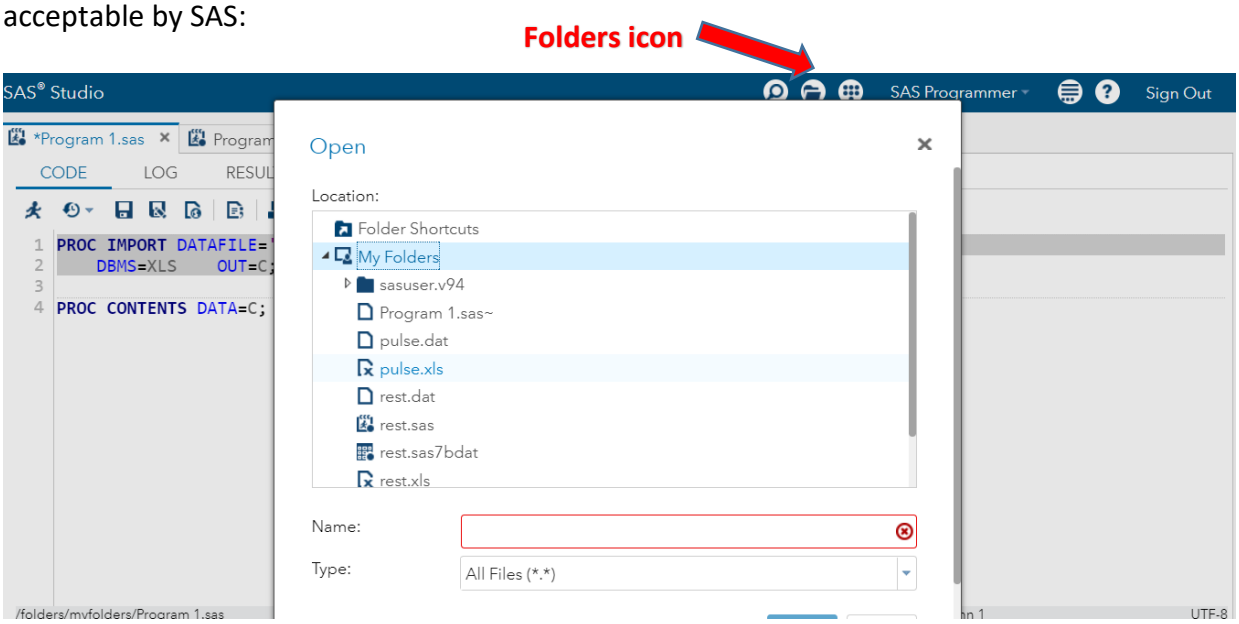


3. Using the shared folder.

Now you can save your SAS codes and data files in the shared folder or its subfolders. SAS knows the newly created folder as /folders/myfolders/. For example, you can read data with proc import

```
PROC IMPORT DATAFILE='/folders/myfolders/pulse.xls'
DBMS=XLS        OUT=C;     GETNAMES=YES;        run;
```

Or you can click on the folders icon within SAS Studio and click on the data file, in any format acceptable by SAS:



A code for importing that clicked dataset will appear. It is already executable, you can hit F3 to run the code and read the data.

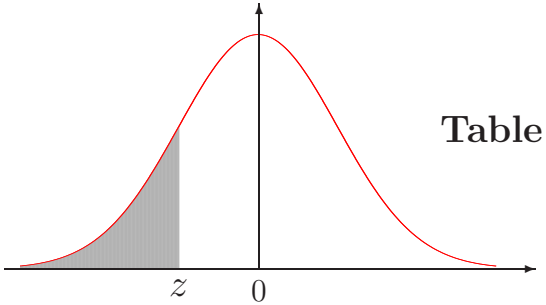
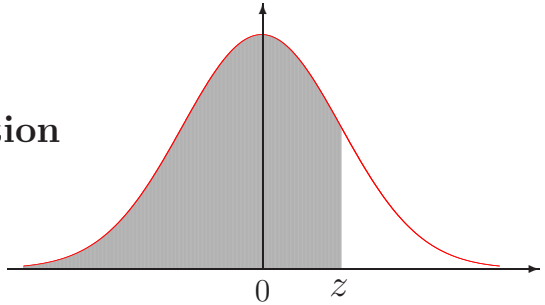


Table of the Standard Normal Distribution

$$\boldsymbol{P}(Z \leq z)$$



z	-0.09	-0.08	-0.07	-0.06	-0.05	-0.04	-0.03	-0.02	-0.01	-0.00
-3.9	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
-3.8	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.7	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.6	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0002	.0002
-3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
-3.4	.0002	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003
-3.3	.0003	.0004	.0004	.0004	.0004	.0004	.0004	.0005	.0005	.0005
-3.2	.0005	.0005	.0005	.0006	.0006	.0006	.0006	.0006	.0007	.0007
-3.1	.0007	.0007	.0008	.0008	.0008	.0008	.0009	.0009	.0009	.0010
-3.0	.0010	.0010	.0011	.0011	.0011	.0012	.0012	.0013	.0013	.0013
-2.9	.0014	.0014	.0015	.0015	.0016	.0016	.0017	.0018	.0018	.0019
-2.8	.0019	.0020	.0021	.0021	.0022	.0023	.0023	.0024	.0025	.0026
-2.7	.0026	.0027	.0028	.0029	.0030	.0031	.0032	.0033	.0034	.0035
-2.6	.0036	.0037	.0038	.0039	.0040	.0041	.0043	.0044	.0045	.0047
-2.5	.0048	.0049	.0051	.0052	.0054	.0055	.0057	.0059	.0060	.0062
-2.4	.0064	.0066	.0068	.0069	.0071	.0073	.0075	.0078	.0080	.0082
-2.3	.0084	.0087	.0089	.0091	.0094	.0096	.0099	.0102	.0104	.0107
-2.2	.0110	.0113	.0116	.0119	.0122	.0125	.0129	.0132	.0136	.0139
-2.1	.0143	.0146	.0150	.0154	.0158	.0162	.0166	.0170	.0174	.0179
-2.0	.0183	.0188	.0192	.0197	.0202	.0207	.0212	.0217	.0222	.0228
-1.9	.0233	.0239	.0244	.0250	.0256	.0262	.0268	.0274	.0281	.0287
-1.8	.0294	.0301	.0307	.0314	.0322	.0329	.0336	.0344	.0351	.0359
-1.7	.0367	.0375	.0384	.0392	.0401	.0409	.0418	.0427	.0436	.0446
-1.6	.0455	.0465	.0475	.0485	.0495	.0505	.0516	.0526	.0537	.0548
-1.5	.0559	.0571	.0582	.0594	.0606	.0618	.0630	.0643	.0655	.0668
-1.4	.0681	.0694	.0708	.0721	.0735	.0749	.0764	.0778	.0793	.0808
-1.3	.0823	.0838	.0853	.0869	.0885	.0901	.0918	.0934	.0951	.0968
-1.2	.0985	.1003	.1020	.1038	.1056	.1075	.1093	.1112	.1131	.1151
-1.1	.1170	.1190	.1210	.1230	.1251	.1271	.1292	.1314	.1335	.1357
-1.0	.1379	.1401	.1423	.1446	.1469	.1492	.1515	.1539	.1562	.1587
-0.9	.1611	.1635	.1660	.1685	.1711	.1736	.1762	.1788	.1814	.1841
-0.8	.1867	.1894	.1922	.1949	.1977	.2005	.2033	.2061	.2090	.2119
-0.7	.2148	.2177	.2206	.2236	.2266	.2296	.2327	.2358	.2389	.2420
-0.6	.2451	.2483	.2514	.2546	.2578	.2611	.2643	.2676	.2709	.2743
-0.5	.2776	.2810	.2843	.2877	.2912	.2946	.2981	.3015	.3050	.3085
-0.4	.3121	.3156	.3192	.3228	.3264	.3300	.3336	.3372	.3409	.3446
-0.3	.3483	.3520	.3557	.3594	.3632	.3669	.3707	.3745	.3783	.3821
-0.2	.3859	.3897	.3936	.3974	.4013	.4052	.4090	.4129	.4168	.4207
-0.1	.4247	.4286	.4325	.4364	.4404	.4443	.4483	.4522	.4562	.4602
-0.0	.4641	.4681	.4721	.4761	.4801	.4840	.4880	.4920	.4960	.5000

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998
3.6	.9998	.9998	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999
3.7	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999
3.8	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999
3.9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

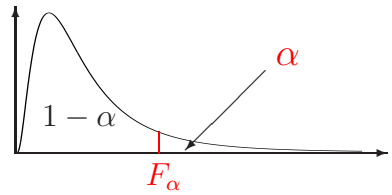
Table of T-Distribution

Critical values t_α for the given right-tail probability α

degrees of freedom	α, the right-tail probability									
	.10	.05	.025	.02	.01	.005	.0025	.001	.0005	.0001
1	3.078	6.314	12.706	15.89	31.82	63.66	127.3	318.3	636.6	3185
2	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60	70.71
3	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92	22.20
4	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610	13.04
5	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.894	6.869	9.676
6	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959	8.023
7	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408	7.064
8	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041	6.442
9	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781	6.009
10	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587	5.694
11	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437	5.453
12	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318	5.263
13	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221	5.111
14	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140	4.985
15	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073	4.880
16	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015	4.790
17	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965	4.715
18	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922	4.648
19	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883	4.590
20	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850	4.539
22	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792	4.452
24	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745	4.382
26	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707	4.324
28	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674	4.276
30	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646	4.234
32	1.309	1.694	2.037	2.141	2.449	2.738	3.015	3.365	3.622	4.198
34	1.307	1.691	2.032	2.136	2.441	2.728	3.002	3.348	3.601	4.168
36	1.306	1.688	2.028	2.131	2.434	2.719	2.990	3.333	3.582	4.140
38	1.304	1.686	2.024	2.127	2.429	2.712	2.980	3.319	3.566	4.115
40	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551	4.094
45	1.301	1.679	2.014	2.115	2.412	2.690	2.952	3.281	3.520	4.049
50	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496	4.014
55	1.297	1.673	2.004	2.104	2.396	2.668	2.925	3.245	3.476	3.985
60	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460	3.962
70	1.294	1.667	1.994	2.093	2.381	2.648	2.899	3.211	3.435	3.926
80	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416	3.899
90	1.291	1.662	1.987	2.084	2.368	2.632	2.878	3.183	3.402	3.878
100	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390	3.861
200	1.286	1.653	1.972	2.067	2.345	2.601	2.838	3.131	3.340	3.789
∞	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.290	3.719

Table of F-distribution.

F_α; critical values such that $\boldsymbol{P}\{F > F_\alpha\} = \alpha$



ν_2, denom. d.f.	α	ν_1, numerator degrees of freedom									
		1	2	3	4	5	6	7	8	9	10
	0.25	2.57	3	3.15	3.23	3.28	3.31	3.34	3.35	3.37	3.38
	0.1	8.53	9	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39
2	0.05	18.5	19	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4
	0.025	38.5	39	39.2	39.2	39.3	39.3	39.4	39.4	39.4	39.4
	0.01	98.5	99	99.2	99.2	99.3	99.3	99.4	99.4	99.4	99.4
	0.005	199	199	199	199	199	199	199	199	199	199
	0.001	999	999	999	999	999	999	999	999	999	999
	0.25	2.02	2.28	2.36	2.39	2.41	2.42	2.43	2.44	2.44	2.44
	0.1	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23
3	0.05	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
	0.025	17.4	16	15.4	15.1	14.9	14.7	14.6	14.5	14.5	14.4
	0.01	34.1	30.8	29.5	28.7	28.2	27.9	27.7	27.5	27.3	27.2
	0.005	55.6	49.8	47.5	46.2	45.4	44.8	44.4	44.1	43.9	43.7
	0.001	167	149	141	137	135	133	132	131	130	129
	0.25	1.81	2	2.05	2.06	2.07	2.08	2.08	2.08	2.08	2.08
	0.1	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92
4	0.05	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6	5.96
	0.025	12.2	10.6	9.98	9.6	9.36	9.2	9.07	8.98	8.9	8.84
	0.01	21.2	18	16.7	16	15.5	15.2	15	14.8	14.7	14.5
	0.005	31.3	26.3	24.3	23.2	22.5	22	21.6	21.4	21.1	21
	0.001	74.1	61.2	56.2	53.4	51.7	50.5	49.7	49	48.5	48.1
	0.25	1.69	1.85	1.88	1.89	1.89	1.89	1.89	1.89	1.89	1.89
	0.1	4.06	3.78	3.62	3.52	3.45	3.4	3.37	3.34	3.32	3.3
5	0.05	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
	0.025	10	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62
	0.01	16.3	13.3	12.1	11.4	11	10.7	10.5	10.3	10.2	10.1
	0.005	22.8	18.3	16.5	15.6	14.9	14.5	14.2	14	13.8	13.6
	0.001	47.2	37.1	33.2	31.1	29.8	28.8	28.2	27.6	27.2	26.9
	0.25	1.49	1.6	1.6	1.59	1.59	1.58	1.57	1.56	1.56	1.55
	0.1	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32
10	0.05	4.96	4.1	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
	0.025	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72
	0.01	10	7.56	6.55	5.99	5.64	5.39	5.2	5.06	4.94	4.85
	0.005	12.8	9.43	8.08	7.34	6.87	6.54	6.3	6.12	5.97	5.85
	0.001	21	14.9	12.6	11.3	10.5	9.93	9.52	9.2	8.96	8.75
	0.25	1.4	1.49	1.48	1.47	1.45	1.44	1.43	1.42	1.41	1.4
	0.1	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2	1.96	1.94
20	0.05	4.35	3.49	3.1	2.87	2.71	2.6	2.51	2.45	2.39	2.35
	0.025	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77
	0.01	8.1	5.85	4.94	4.43	4.1	3.87	3.7	3.56	3.46	3.37
	0.005	9.94	6.99	5.82	5.17	4.76	4.47	4.26	4.09	3.96	3.85
	0.001	14.8	9.95	8.1	7.1	6.46	6.02	5.69	5.44	5.24	5.08